

EFFECT OF SOIL ENVIRONMENT ON CROP CULTIVATION

Lesson 03 — Comprehensive Lecture & Study Notes

1. Soil Formation and Soil Profile Development

Soil Definition: Soil is a dynamic body composed of minerals, organic matter, various living organisms, air, and water, which serves as a natural medium for terrestrial plant growth.

Soil Formation Process

Rocks → *Weathering (Physical, Chemical & Biological)* → **Parent Material** → *Pedogenesis (Further Weathering & Development)* → **Soil**

Rocks: Dense solid masses formed by large aggregates of minerals. Classified based on their mode of origin:

- **Igneous Rocks:** Formed when molten magma cools and solidifies on or near the Earth's surface. *Example: Granite.*
- **Sedimentary Rocks:** Formed by the accumulation, compaction, and cementation of mineral particles and eroded sediments transported by wind or water. *Example: Limestone.*
- **Metamorphic Rocks:** Formed when pre-existing igneous or sedimentary rocks undergo profound physical/chemical changes due to high heat and intense pressure underground. *Example: Marble.*

Soil Profile

A soil profile is the vertical section of soil from the surface down to the parent rock, displaying distinct layers called **horizons** formed through pedogenesis (soil generation).

Horizon	Description / Characteristics
O - Organic Horizon	Contains decomposed and decomposing organic matter. • A_{00}: Un-decomposed leaf litter. • A_0: Humified mineral/organic matter (initial stage of decomposition).
A - Eluvial / Topsoil Horizon	Zone of leaching (eluvial zone). Minerals are added to the soil as leached organic matter decomposes. • A_1: Rich in minerals and organic matter. • A_2: Contains leached nutrients.
B - Illuvial / Subsoil Horizon	Zone of accumulation (enriched layer) where nutrients leached from Horizon A deposit. • B_1: Contains nutrients leached down from Horizon A. • B_2: Primary layer where leached nutrients accumulate. • B_3: Transition zone transforming parent material into soil.
C - Parent Material Horizon	Unconsolidated parent material derived from weathered rock.
R - Bedrock Horizon	Solid, unweathered parent bedrock.

Notes on Soil Profiles:

- In undisturbed soils, these layers are clearly visible. However, in agricultural lands, topsoil layers are often mixed by tillage.
- In hyper-evaporative regions, horizons may become indistinct due to salts evaporating upwards with capillary water.
- In heavy rainfall regions, Horizon A can be extensively washed away/leached.

Factors Affecting Soil Profile Development

Distinct soil profiles develop due to key processes: addition and decomposition of surface organic matter, vertical percolation of water, and mineral release via weathering. The 5 primary factors are:

1. **Parent Material:** Determines mineral composition and texture.
2. **Climate:** Temperature and precipitation dictate weathering rates and leaching.
3. **Organisms (Biota):** Mix soil, decompose organic matter, and recycle nutrients.
4. **Topography (Relief):** Influences runoff, erosion, and water drainage.
5. **Time:** Soil development progresses over thousands of years.

Chronological Stages of Soil Profile Development

1. Initial weathering of rock begins.
2. **~10 Years:** Parent material forms (Horizon C appears).
3. **~100 Years:** Leaf litter accumulates and decomposes to form organic matter (Horizon A forms).
4. **~1,000 Years:** Horizon A thickens, Horizon O fully develops, and Horizon B forms via eluvial leaching from A.
5. **~10,000 Years:** Advanced weathering produces iron oxides and clay minerals that leach downward, enriching the soil profile.

6. **~100,000 Years:** An E horizon forms above Horizon B (rarely observed in Sri Lanka).

Agricultural Importance of Soil Profile and Depth

- Enables determination of effective total soil depth and thickness of individual layers.
- Guides crop selection based on root depth requirements.
- Helps determine required tillage depth and selection of land preparation equipment.
- Provides insights into nutrient supply across different layers and soil drainage conditions.

Soil Depth Classification & Crop Suitability

- **Shallow Soil (< 50 cm):** Suitable for shallow-rooted crops (e.g., vegetables, chillies, maize).
- **Moderately Deep Soil (50 cm – 100 cm):** Suitable for medium-rooted crops.
- **Deep Soil (100 cm – 150 cm):** Suitable for deep-rooted crops (e.g., mango, timber trees). Excellent drainage.
- **Very Deep Soil (> 150 cm):** Ideal for extensive root systems. Note: Very deep sandy soils may suffer from high nutrient leaching.

2. Soil Components, Influence, and Management

Soil consists of solid materials (minerals and organic matter), soil air, soil water, and soil organisms.

Soil Minerals

Homogeneous inorganic substances with defined chemical compositions. Derived from primary and secondary minerals:

- **Primary Minerals:** Derived directly from igneous rocks; abundant in the sand fraction. Highly resistant to weathering. Silicate-rich (e.g., Quartz, Feldspar).
- **Secondary Minerals:** Derived from weathering of sedimentary/metamorphic rocks or primary minerals. Include layer silicates (e.g., Kaolinite).

Clay Minerals

Secondary aluminosilicate minerals with layer structures. Highly chemically active, crystalline, negatively charged (capable of cation adsorption/exchange), and capable of swelling upon water absorption between lattice units.

- **1:1 Type (Silica : Alumina = 1:1):** Example: *Kaolinite* (non-expanding, lower CEC).
- **2:1 Type (Silica : Alumina = 2:1):** Example: *Montmorillonite* (expanding, high CEC).
- **Non-silicate Minerals:** Calcite, Hematite, Allophane.

Soil Organic Matter & Humus

A heterogeneous mixture produced by microbial decomposition of plant and animal residues. Decomposed organic matter forms **Humus** (comprising humic and non-humic substances).

Agricultural Importance of Organic Matter:

- **Nutrient Supply:** Acts as a nutrient reservoir; humus adsorbs and exchanges cations.
- **Physical Properties:** Improves soil structure, reduces erosion, prevents structural breakdown during tillage, increases water holding capacity, and acts as a pH buffer.
- **Biological Function:** Provides energy and carbon source for soil microbes; stimulates plant root growth.

Soil Organisms

- **By Size:**
 - *Macro-organisms* (> 2 cm): Earthworms, rodents, snails.
 - *Meso-organisms* (0.2 cm – 2 cm): Mites, lichens.
 - *Micro-organisms* (< 0.2 cm): Bacteria, fungi, algae, protozoa.
- **By Ecological/Functional Role:** Plant-feeders (nematodes), detritivores, predators, parasites, fungal/bacterial feeders.
- **By Metabolic Energy Source:** Autotrophs (algae, plants) and Heterotrophs (most bacteria/fungi).

Important Soil Bacteria

- **Ammonifying Bacteria:** Decompose organic nitrogen into amino acids and further down to ammonia/ammonium ions (NH_4^+).
- **Nitrifying Bacteria:** Oxidize ammonia/ammonium into plant-available nitrate ions (NO_3^-) under aerobic conditions.
- **Denitrifying Bacteria:** Convert nitrates into atmospheric nitrogen gas under anaerobic (waterlogged) conditions.

Soil Aeration

The exchange of soil air with atmospheric air (gas circulation in soil pore space).

- **Factors Affecting Soil Aeration:** Soil water content (inverse relationship), total porosity, texture/structure, bulk density (higher density reduces air capacity), soil depth (lower air in deep layers), crop root respiration, and microbial activity.
- **Importance:** Essential for root and plant growth, microbial activity, preventing toxic gas accumulation, active water/nutrient absorption, and seed germination. Poor aeration promotes denitrification and root diseases.

Soil Water

Water retained in soil pore spaces and around soil particles.

Retentive Forces of Soil Water

- **Cohesion & Adhesion:** Cohesion (attraction between like water molecules); Adhesion (attraction between water molecules and solid soil particles).
- **Capillarity:** Rise of water through narrow capillary pores against gravity.
- **Surface Tension & Polarity:** Water molecules are polar (H^+ , OH^-), allowing electrostatic binding around clay micelles.

pF Value: The logarithm of the suction force (water column height in cm) required to extract water from the soil.

Classification of Soil Water

Classification Basis	Water Category	Characteristics & Availability
Physical Classification	Gravitational Water	Loosely held in macropores; drains freely under gravity. Unavailable to plants. (pF = 0 to 2.5)
	Capillary Water	Held in micro/capillary pores by cohesive/adhesive forces. Readily available to plants. (pF 2.5 to 4.2)
	Hygroscopic Water	Tightly bound as a thin film around soil particles by strong adhesive forces. Removed only by oven-drying. Unavailable to plants. (> pF 4.2)
Biological Classification	Available Water	Water retained between Field Capacity (pF 2.5) and Permanent Wilting Point (pF 4.2).
	Unavailable Water	Includes hygroscopic water and tightly held capillary water (> pF 4.2).
	Superfluous Water	Excess gravitational water present in macro-pores above field capacity.

Key Moisture Constants

- **Field Capacity (FC):** Moisture content remaining in soil after excess gravitational water has fully drained following saturation. (pF = 2.5). Macro-pores are filled with air; micro-pores are filled with water.
- **Permanent Wilting Point (PWP):** Soil moisture percentage at which plants wilt irreversibly and cannot recover even when placed in a humid environment or watered. (pF > 4.2).

Determination Procedures

1. Determination of Field Capacity (FC)

Procedure: Take a soil sample using a 10 cm galvanized cylinder. Place on filter-paper-wrapped wooden block in a water dish, cover top with polythene. Weigh after 3 days (m_3). Transfer to evaporating dish, weigh wet (w_2), then oven-dry at 105°C to constant weight (w_3).

$$\text{Field Capacity (\%)} = [(m_3 - m_1) - (w_3 - w_1)] / (w_3 - w_1) \times 100$$

2. Determination of Permanent Wilting Point (PWP)

Procedure: Plant sunflower seeds in 500g soil in a perforated pot. Allow growth to 3 leaf pairs. Stop watering, seal pot, allow plant to wilt. Place in a bell jar with water vessel under black polythene cover. If plant recovers next day, expose to air for 2 hours and repeat until permanent wilting occurs. Measure soil moisture content gravimetrically.

3. Soil Moisture Measurement Methods

- **Gravimetric Method:** Sample wet weight vs. oven-dry weight (105°C).

$$\text{Moisture \%} = [(\text{Wet Weight} - \text{Dry Weight}) / \text{Dry Weight}] \times 100$$

- **Gypsum Block Method:** Measures electrical conductivity between embedded electrodes, which correlates with soil moisture tension.

- **Tensiometer Method:** Measures soil water matric potential directly in the field via a porous ceramic cup connected to a vacuum gauge.

3. Soil Texture, Structure, and Physical Properties

Soil Texture

The relative percentage proportions of sand, silt, and clay particle size fractions in a soil mass.

ISSS Particle Category	Particle Diameter Range (mm)
Gravel / Stones	> 2.0 mm
Coarse Sand	2.0 – 0.2 mm
Fine Sand	0.2 – 0.02 mm
Silt	0.02 – 0.002 mm
Clay	< 0.002 mm

Importance of Soil Texture: Essential for selecting appropriate crops, tillage tools, irrigation methods and intervals, nursery bed types, fertilizer application rates/frequency, and soil conservation practices.

Determination of Soil Texture (Pipette Method & Textural Triangle)

Steps: Destroy organic matter with H_2O_2 ; disperse using $NaOH$; sieve coarse sand (0.02mm); transfer to 1000 ml cylinder. Sample aliquots using 25 ml pipette at specific depth (10cm) and settling times (6 hrs 10 mins for clay; 3 mins 42 secs for silt+clay). Oven-dry at 105°C and calculate percentages:

$$\text{Silt + Clay \%} = [(w_2 - w_1) \times 40] / 40g \times 100 = x\%$$

$$\text{Clay \%} = [(w_4 - w_3) \times 40] / 40g \times 100 = y\%$$

$$\text{Silt \%} = x - y \quad | \quad \text{Sand \%} = 100 - x$$

Plot these percentages on the **USDA / ISSS Textural Triangle** by drawing lines parallel to triangle sides to pinpoint the exact soil textural class (e.g., Sandy Loam, Clay Loam, Silt Loam).

Field Feel / Roll Method

- Moisten 50–100g soil and attempt to roll between palms.
- Cannot form a ball → **Sand**
- Forms a ball but breaks upon rolling → **Sandy Loam**
- Rolls into a cylinder but cracks when bent into a ring → **Clay Loam**
- Rolls smoothly into a ring without cracking → **Clay**

Soil Consistency

Resistance of soil to deformation or rupture under applied mechanical force. Expressed as soil firmness (hard, friable, loose), cohesion, plasticity, and stickiness. Important for root penetration and determining soil workability windows.

Soil Structure

The arrangement of primary soil particles (sand, silt, clay) into aggregated secondary units called **peds**.

- **Structural Types (Shapes):** Granular (small spherical peds common in topsoil/root zones), Blocky (angular/sub-angular), Prismatic (vertical pillars), Columnar, Platy (flat horizontal sheets), Single Grain (non-aggregated sand), Massive.
- **Grades/Classes:** Weak, Moderate, Strong; Fine, Medium, Coarse.
- **Management:** Destroyed by excess sodium (Na^+ dispersion) or heavy machinery; improved by organic matter additions, proper drainage, and cover crops.

Soil Color

Determined using the **Munsell Color Chart** (Hue, Value, Chroma) under both wet and dry conditions.

Soil Color	Associated Soil Condition & Properties
Black / Dark Brown	High organic matter content; rich fertility; often restricted drainage.
Red	Good drainage and aeration; high unhydrated iron oxides (Fe_2O_3); low denitrification.
Yellow	Hydrated iron oxides; moderate to poor drainage conditions.
Light Grey / Pale	Leached mineral horizons; prolonged waterlogging/gleying; intense leaching.
White	High accumulation of salts, carbonates, or pure quartz sand.

Soil Density and Porosity

1. Bulk Density (D_b)

Mass of dry soil per unit total bulk volume (including solids and pore spaces).

$$\text{Bulk Density } (D_b) = \text{Dry Soil Mass (g)} / \text{Total Volume (cm}^3\text{)} = (w_2 - w_1) / (\pi r^2 h)$$

2. Particle Density (D_p)

Mass of dry soil per unit volume of solid particles only (excluding pore space). Standard average $\sim 2.65 \text{ g/cm}^3$.

$$\text{Particle Density } (D_p) = (m_2 - m_1) / [(m_4 - m_1) - (m_3 - m_2)]$$

3. Soil Porosity

Percentage of total soil volume occupied by pore spaces (macro and micro pores).

$$\text{Porosity (\%)} = [1 - (\text{Bulk Density} / \text{Particle Density})] \times 100\%$$

4. Chemical Properties of Soil

Soil Colloids

Extremely small particles (< 0.002 mm diameter) with high specific surface area, carrying electrostatic charges. Divided into:

- **Inorganic Colloids (Clay Colloids):** Crystalline aluminosilicates, stable structure, permanent/variable negative charge.
- **Organic Colloids (Humus Colloids):** Amorphous, non-crystalline, temporary/dynamic, derived from amino acids and organic acids, extremely high cation exchange capacity.

Soil pH & Reaction

Soil pH is the negative logarithm of active hydrogen ion (H^+) concentration in the soil solution:

$$pH = -\log_{10}[H^+] \quad | \quad pH < 7 \text{ (Acidic)} \quad | \quad pH = 7 \text{ (Neutral)} \quad | \quad pH > 7 \text{ (Basic/Alkaline)}$$

Measurement: Measured via pH meters (glass electrode), pH indicator papers, or colorimetric indicators.

Soil Acidity

- **Causes:** High rainfall leaching basic cations (Ca^{2+} , Mg^{2+} , K^+ , Na^+); organic matter decomposition forming organic acids under cool conditions; prolonged crop extraction; long-term use of acidifying nitrogen fertilizers (e.g., Ammonium Sulfate); acid rain.
- **Agricultural Problems:** Aluminum (Al^{3+}) and Manganese (Mn^{2+}) toxicity; Phosphorus fixation; reduced availability of *Ca*, *Mg*, *Mo*; suppressed microbial activity.

Soil Salinity and Alkalinity

- **Saline Soils:** High concentration of soluble salts in soil solution. Plants suffer from osmotic stress (exosmosis), leading to physiological drought and death. (pH usually 7.5 – 8.5).
- **Alkaline / Sodic Soils:** High Exchangeable Sodium Percentage (ESP > 15%). Poor physical structure, dispersed clay, extremely poor drainage/aeration.
- **Remediation:** Application of Gypsum ($CaSO_4 \cdot 2H_2O$) for sodic soils; flushing/leaching excess salts with good-quality freshwater; surface soil scraping; installing subsurface drainage lines.

Ion Exchange Phenomena

- **Cation Exchange Capacity (CEC):** Total quantity of exchangeable cations that a soil can adsorb per unit dry weight. Critical for nutrient retention and fertility. Driven by ionic charge, ionic radius, and ion concentration in solution.
$$\text{Base Saturation (\%)} = [\text{Sum of Exchangeable Basic Cations } (Ca^{2+}, Mg^{2+}, K^+, Na^+) / \text{Total CEC}] \times 100\%$$
- **Anion Exchange Capacity (AEC):** Adsorption and exchange of anions (NO_3^- , SO_4^{2-} , $H_2PO_4^-$) on positively charged colloidal sites, predominant in highly weathered acidic soils.
- **Electrical Conductivity (EC):** Measured with an EC meter to assess total soluble salt concentration in soil solution.

5. Major Soil Groups of Sri Lanka and Agricultural Potential

Based on the Panabokke-Moormann soil classification system, Sri Lanka is divided into 14 prominent Great Soil Groups:

Soil Group	Distribution / Location	Key Characteristics	Suitable Crops
Red Red-Brown Earths (RBE)	Dry Zone (Vavuniya, Hambantota, Polonnaruwa, Anuradhapura)	Neutral pH, good depth and drainage, deficient in organic matter and nitrogen. Highly fertile when irrigated.	Cereals, pulses, chillies, vegetables, onions, field crops.
Red Yellow Podzolic (RYP)	Wet Zone (Ratnapura, Kalutara, Kegalle, Matara, Kurunegala, Kandy)	Acidic soils, severe leaching, low Ca and Mg. Organic matter high in lowlands, low in highlands.	Tea, rubber, coconut, fruits, spices, export tree crops.
Non-Calcic Brown (NCB)	Dry/Intermediate Zone (Ampara, Batticaloa, Gal Oya Valley)	Low organic matter, rich in Ca and Mg, poor unstable structural aggregation.	Pulses, chillies, vegetables, field crops, pasture.
Red Yellow Latosols	Dry Zone coastal strips (North, East, North-West)	Slightly acidic, deep, sandy texture, low CEC, high infiltration capacity.	Cash crops, deep-rooted perennials, citrus, cashew.
Reddish Brown Latosolic	Intermediate Zone (Kandy, Matale, Kegalle)	Moderate texture, good CEC, slightly acidic, low in N and organic matter.	Cocoa, spice crops, coffee, perennial fruit trees.
Immature Brown Loams	Wet & Intermediate hilly areas	Young, weakly weathered soils rich in primary minerals.	Export agricultural crops, vegetables.
Regosols & Alluvial Soils	Coastal belts (Regosols) & River basins (Alluvial)	Regosols are coarse sandy; Alluvial soils are highly variable, fertile, regularly deposited by floodwaters.	Coconut (Regosols); Paddy, vegetables, sugarcane (Alluvial).

6. Soil Degradation and Erosion

Soil Degradation: The decline in soil quality and biological productivity caused by adverse human activities and environmental stress.

Soil Erosion Process

The detachment, transportation, and deposition of soil particles from their original location by erosive agents (water, wind, gravity).

1. **Detachment:** Separation of soil particles caused by raindrop impact, overland flow, or mechanical tillage.
2. **Transportation:** Movement of detached particles by runoff water, wind, or wave action.
3. **Deposition:** Settling of transported sediments in low-lying areas, reservoirs, or riverbeds.

Types of Water Erosion

- **Splash Erosion:** Direct impact of raindrops shattering soil aggregates and splashing soil particles.
- **Sheet Erosion:** Uniform removal of a thin surface layer of soil by shallow overland water flow.
- **Rill Erosion:** Small, well-defined channels cut into topsoil by concentrated runoff.

- **Gully Erosion:** Advanced rill erosion forming deep, wide channels that cannot be smoothed by normal tillage.
- **Stream Bank Erosion:** Washing away of riverbanks by flowing water currents.
- **Landslides & Mass Wasting:** Downslope movement of entire soil masses under gravitational pull.

Universal Soil Loss Equation (USLE)

Used to estimate average annual soil loss from agricultural fields:

$$X = R \times K \times L \times S \times C \times P$$

- **X:** Estimated average annual soil loss.
- **R:** Rainfall erosivity factor.
- **K:** Soil erodibility factor.
- **L:** Slope length factor.
- **S:** Slope steepness factor.
- **C:** Crop management / cover factor.
- **P:** Supporting conservation practice factor.

7. Soil Conservation and Rehabilitation Methods

Soil conservation involves managing soil to prevent erosion, maintain fertility, and ensure sustainable agricultural productivity.

1. Mechanical / Engineering Measures

- **Contour Drains:** Channels excavated along contour lines to intercept and safely discharge excess runoff. Includes main drains and lateral drains (block drains, silt-trap drains).
- **Contour Bunds:** Earth ridges constructed across slopes along contour lines to reduce runoff velocity and promote infiltration.
- **Terracing:** Converting steep sloped land into stepped flat platforms (Bench terraces, Sloping terraces, Ridge terraces).
- **Stone Terraces / Stone Walls:** Constructing stone barriers perpendicular to land slope using local stones.

2. Agricultural / Agronomic Measures

- **Proper Land Use & Tillage:** Plowing along contour lines; zero/minimum tillage; plowing during dry spells to minimize erosion.
- **Mulching:** Covering bare soil surface with crop residues, straw, sawdust, or synthetic polythene to absorb raindrop impact and conserve moisture.
- **Intercropping & Cover Cropping:** Planting ground cover legumes or mixed crops to maintain dense canopy cover.

3. Biological Measures

- **Strip Cropping:** Alternating strips of erosion-permitting crops (e.g., maize) with dense erosion-resisting crops (e.g., legumes) along contours.
- **Vetiver Grass Hedgerows:** Planting deep-rooted, non-invasive Vetiver grass (*Vetiveria zizanioides*) along contours to trap sediment and slow runoff.

- **Sloping Agricultural Land Technology (SALT):** Double rows of nitrogen-fixing hedgerow trees (e.g., *Gliricidia*, *Leucaena*) planted along contours, pruned regularly to provide mulch and nitrogen enrichment.

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